Unit 1

Reading 1:

Will MOOCs Make College Obsolete?

Sara Elliott

1 If you have an online degree, or are pursuing one, you may be seeing a shift in the way long distance learning is perceived by employers and the public in general. Early resistance to online education may have been caused by concern about diploma mills or worry about a lack of oversight in the virtual classroom. Those notions and other ideas about using computers and the Internet to educate students may soon be up for revision, too.

2 Massive open online courses (MOOCs) are making waves in academia, and although it's too soon to determine their ultimate impact on the ivy-covered halls of universities across the country and the world, they are inspiring a lively debate among educators and students.

3 The first MOOC arrived on the scene back in 2011 when two Stanford professors offered their artificial intelligence class online — for free. Enrollment topped 160,000, and the unprecedented success of the experiment inspired other schools, many of them top tier universities, to mount their own offerings.

4 On the pro side of the MOOC debate, proponents believe making quality educational instruction available to the masses will provide higher quality and lower cost for many, and might easily be the best instructional model for the future. MOOCs can be attended by thousands or even hundreds of thousands of students at a time. In fact, Sebastian Thrun, one of the Stanford professors who started it all, sees a day in which as few as 10 universities fulfill the world's higher education needs.

5 That may be an extreme view, but a number of companies are springing up to take MOOC learning to the next level, including Coursera, Udacity and EdX. Many big universities are participating in MOOCs to one degree or another, too. Some examples are Stanford, Princeton, the University of California Berkeley and Duke University.

6 Where MOOCs may provide quality instruction for many, opponents often cite that very fact as its biggest failing. MOOCs don't offer any individualized training, much less tutoring or mentoring, and some educators feel adopting it as a for-credit standard in the future will leave some students behind and alienate others. A common argument is that individualized, face-to-face teaching works best in many cases because no single instructional technique will fulfill the needs of all students. The lively exchange of ideas that can electrify a classroom is the province of the brick-and-mortar school, not the virtual classroom.

7 Some find other aspects of MOOC instruction troubling. In the first MOOCs, plagiarism was a problem and overall completion rates were low. This may have been because some students were curious about the format rather than focused on scholarship. Still, for MOOCs to be widely offered for credit, they will have to be structured to provide oversight, consistency and student accountability.

8 How will MOOCs fare in the future? No one knows for sure, but the ultimate modern classroom may turn out to be a blend of the old and the new, with on-campus activities like labs, debates and other interactive engagements working in concert with massive online offerings to provide a comprehensive, flexible learning experience that's cost-effective and convenient.

Language Enhancement

1. The underlined words in the following sentences are either active words or academic words marked in Academic Word List (AWL). Choose a, b, or c that has the closest meaning to the underlined words.

1) There has been a lot of resistance to tax increases, even those designed to benefit education. (a)
   a. opposition b. fighting c. defense
2) Tuition fees for some top tier universities have reportedly topped $35,000 this year. (a)
   a. been higher than b. done better than c. been in the highest position
3) The popular science association is mounting an exhibition to illustrate how life evolved from water. (c)
   a. climbing b. increasing c. organizing
4) These forms can provide you with easy access to our overseas office to help fulfil your international shipping needs. (c)
   a. accomplish b. carry out c. satisfy
5) The candidate's political views were too extreme and met with resistance from citizens. (c)
   a. very great in degree b. very serious c. unacceptable and unreasonable
6) Davis has earned a reputation as an outspoken opponent of any kind of nuclear waste dumping (倾倒) at sea. (c)
   a. enemy b. competitor c. a person who disagrees
7) These teaching quality standards have been adopted by many colleges and universities; others have established their own standards. (b)
   a. approved by voting b. accepted and used c. taken in and brought up
8) My American tutor in college advised me to take all the required credits in the first two years. (c)
   a. praises b. loans c. courses

2. A preposition can be very useful in forming a collocation.

A. Scan Reading 1 and fill in the blanks with the prepositions used in the collocations.

1) Early resistance to online education (Para. 1)
2) concern about diploma mills (Para. 1)
3) a lack of oversight (Para. 1)
4) be up for revision (Para. 1)
5) their ultimate impact on the ivy-covered halls (Para. 2)
6) On the pro side of the MOOC debate (Para. 4)
7) making quality educational instruction available to the masses (Para. 4)
8) participating in MOOCs to one degree or another (Para. 5)
9) focused on scholarship (Para. 7)
10) working in concert with massive online offerings (Para. 8)

B. Complete the sentences with the collocations from Step A. Change the form where necessary.

1) The management is working in concert with the staff to upgrade the quality of their products.
2) However, many companies concentrate on the technological aspects and neglect to focus on the "human side" of the product marketing.
3) I wonder if the agency has any idea how much wildlife has been killed by the lack of food and water.
4) Whether there is any truth in the article is up for discussion but we'd certainly be interested in reading your views on the matter.
5) Professor Dodge will act as a mentor — he'll be available to the tutors with lesson plans or suggestions for activities.

3. Study the word building skill in the box and work on the tasks.

A. Write down the clipped form of the original word in the following table. The first one has been done as an example. Consult the dictionary where necessary.

Original word	Clipped word	Original word	Clipped word
1) gasoline	gas	6) limousine	limo
2) hamburger	burger	7) parachute	chute
3) doctor	doc	8) cockroach	roach
4) gymnasium	gym	9) Internet	net
5) memorandum	memo	10) refrigerator	fridge

B. Write down the original form of the clipped word in the following table. The first one has been done as an example. Consult the dictionary where necessary.

Clipped word	Original word	Clipped word	Original word
1) auto	automobile	6) photo	photograph
2) champ	championship	7) pro	professional
3) dorm	dormitory	8) quake	earthquake
4) grad	graduate	9) bro	brother
5) lab	laboratory	10) vet	veterinarian / veteran

4. Translating

Translate into English the following paragraph, choosing the subject of each sentence carefully.

科技中介学习（technology-mediated learning）迅猛扩展，引发了种种吹捧与担忧，不过绝大多数美国传统的高等教育中心都没能正视这一变化的节奏及其对学生的含意。很有可能根本没有人能跟得上这一改变：科技发展得实在是太快了。无论如何，我们坚持努力。这些发展能否真正为我们达成先进教育的目标服务呢？我们迫切地需要知道答案。

The rapid spread of technology-mediated learning has set off fits of praise and concern — yet the U.S.'s traditional centers of higher education have mostly failed to confront the pace of change and the implications for students. There is probably no way anyone can keep up with this transformation: The technology is simply evolving too rapidly. Nevertheless, we keep trying. Will these developments truly serve our goals for advanced education? We need to know urgently.

Unit 2
Reading 1:

The More Time That Children Chat on Social Media, The Less Happy They Feel

Jamie Doward

1 Perhaps Facebook should carry a health warning. A study has revealed that the children who spend more time on online social networks feel less happy in almost all aspects of their lives.

2 The research by a team of economists at the University of Sheffield, to be presented at this week's Royal Economic Society annual conference in Bristol, shows that the more time children spend chatting on Facebook, Snapchat, WhatsApp and Instagram, the less happy they feel about their school work, the school they attend, their appearance, their family and their life overall. However, they do feel happier about their friendships.

3 Economists found that spending just one hour a day on social networks reduces the probability of a child being completely happy with his or her life overall by around 14%. They found that this was three times as high as the estimated adverse effect on well-being of being in a single-parent household — and larger than the effect of playing truant.

4 The findings are likely to stoke the debate about the upsides and downsides of social media.

5 More than 90% of 16- to 24-year-olds use online social networks and while most sites stipulate a minimum user age of 13, few apply any checks. A BBC survey found that more than three-quarters of 10- to 12-year-olds have social media accounts. A report by the media watchdog Ofcom found that more than half of children aged as young as three and four use a tablet while one in seven has their own device.

6 The amount of time that children between eight and 11 and those aged 12-15 spend online has more than doubled in a decade, the Ofcom report found. Teenagers now spend nearly three and a half more hours a week online than they do watching television.

7 Social networking has altered childhood dramatically in the past decade and is becoming a concern for politicians and organizations responsible for safeguarding children. The NSPCC cited social media as a major cause of the dramatic increase in the numbers of children admitted to hospital after self-harming. The new research, which asked 4,000 10- to 15-year-olds to rate from one to seven how happy they were with different aspects of their lives, reveals that girls are more adversely affected than boys, as online social networking makes them feel less happy about specific areas of their life, in particular about their appearance and the school they attend. Boys were less happy with their friendships.

8 The research suggests that going online makes children more likely to make negative social comparisons with others.

9 "The problem with making comparisons in online media is that people tend to portray themselves in an idealized state," said Philip Powell, one of the economists who conducted the research. "There is evidence that people think other people are happier than them after interacting with them online because we tend to post videos and chat that present this positive image."

10 Powell and his team split the data so that they could compare the effects of going online on children with low and high self-esteem. Unsurprisingly, perhaps, they found the effects were worse for those who lacked confidence.

11 Powell said cyberbullying could be another explanation for links between unhappiness and children's use of social media. "There's evidence the longer young people spend online the more likely they are to be victims of bullying," he said.

12 However, the economists were surprised to find nothing to support the popular theory that time spent on social networks had an adverse effect on children because it left them less time to do other, potentially more rewarding, activities.

13 "Our findings show that social media use can be detrimental on average to young people and this is consistent with a number of findings in previous studies," Powell said. "We can't say any social media is bad but we can say that the more social media children use, the higher the likelihood that they will be dissatisfied with different domains of their life and their life overall."

Language Enhancement:

1. Using appropriate synonyms may help you paraphrase ideas, avoid repetition and add variety to your writing. Write down as many synonyms as possible of the underlined words in the sentences. The first one has been given as an example.

1) Many fear that budget cuts will have an <u>adverse</u> effect on education.
Synonyms: damaging; negative; bad; unfavorable; unpleasant; unhelpful; harmful; undesirable; detrimental

2) The <u>upside</u> of moving would be a shorter commute to work.
Synonyms: advantage; benefit; positive aspect; good thing; plus

3) The <u>downside</u> of the camera is that the batteries have to be replaced often.
Synonyms: disadvantage; drawback; shortcoming; negative aspect; bad thing

4) This one small event <u>altered</u> the course of history.
Synonyms: changed; transformed; modified; adjusted; varied; revised

5) You need to <u>safeguard</u> your computer against viruses.
Synonyms: protect; defend

6) Heavy rains, in the central valley <u>in particular</u>, have driven up the price of vegetables.
Synonyms: especially; specially; particularly; specifically

7) The lawyer <u>portrayed</u> his client as a victim of child abuse.
Synonyms: described; depicted; represented; showed; pictured; rendered; interpreted

8) The movie portrays an <u>idealized</u> version of middle-class life.
Synonyms: perfect; flawless; ideal; faultless; fanciful

9) No <u>previous</u> experience is necessary for the job.
Synonyms: earlier; prior; former; preceding; past

10) Childcare is no longer solely a female <u>domain</u>.
Synonyms: field; area; sphere; realm

2. Word building

A. Write down the structure of the blending words (blends) in the following table. Then translate them into Chinese. The first one has been done as an example. Consult the dictionary where necessary.

Blend	Structure	Chinese meaning
1) smog	smoke + fog	雾霾
2) motel	motor + hotel	汽车旅馆
3) Chinglish	Chinese + English	中式英语
4) hazmat	hazardous + material	危险品
5) telethon	television + marathon	长时间的电视节目
6) webinar	web + seminar	在线研讨会
7) edutainment	education+ entertainment	教育性娱乐；寓教于乐型产品
8) biorhythm	biological + rhythm	生物节律；人体生理周期
9) psywar	psychological + warfare	心理战
10) bookmobile	book + automobile	流动图书馆

B. Complete the sentences with the blends in Step A. Change the form where necessary.

1) Donations raised during the four-hour telethon will go to the Ministry of Civil Affairs, the Red Cross Society of China and the China Charity Federation, according to the director.
2) It is necessary to check the time length and major topic of your webinar, because you are inviting participants from other countries and different time zones.
3) Interrogation (审讯) is a type of psywar between the police questioners and criminal suspects.
4) The biological clock is also called biorhythm. It ties up with people's minds, and controls people's behavior.
5) A(n) hazmat accident can cause great destruction to society and this kind of accident is usually associated with criminal and terrorist activities.
6) If teachers purposefully and flexibly apply the concept of edutainment to classrooms, it will increase the fun of teaching, cover the shortage of textbooks, and have a better effect on learning.

3. Translating:

Translate into English the following paragraph, choosing the predicate verb of each sentence carefully.

手机是一项伟大的发明，但手机似乎也给我们的日常生活造成了一些麻烦。会议室门口通常贴着一条公告：请与会者关闭手机。可是，会议室里手机铃声仍然响成一片。我们都是普通人，并没有多少特别重要的事情，但我们也不会轻易关掉手机。打开手机象征着我们与这个世界的联系。手机在某种程度上反映出我们的“社交饥渴症” (thirst for socialization)。

The cell phone is a great invention, but it seems to have become a bit of a nuisance in our daily lives. There is usually a notice on the door of the meeting room, which reads, "Please turn off your cell phones." However, phones ring now and then during the meeting. Ordinary people like us do not have many important things to handle. Still, we will not switch off our phones easily. Keeping our phones on symbolizes our connection with this world. To a certain extent, the cell phone reflects our "thirst for socialization".

Unit 3
Reading 1

My Laughter Rx for Your Heart

1. One of my favorite moments as a physician occurs when, with a very somber look, I inform patients that there's one thing they absolutely must do in order to make a successful recovery after a cardiac event: Go home and laugh until they cry.

2 You see, we now know that there's far more to maintaining heart health and reversing heart disease than diet, exercise, and cholesterol levels. The latest research indicates that stress, and an inability to deal with it, is a direct contributor to heart disease. For example, a study involving nearly 250,000 people found that anxiety was associated with a 26 percent increase in coronary heart disease over an 11-year period.

3 Anger and hostility rank at the top of the list of heart-harmful emotions. Harvard Medical School researchers recently found that 40 percent of patients who suffered a heart attack reported significant anger within the previous year, and roughly 8 percent of that group reported that they felt rage within two hours of heart attack symptoms.

4. But while studies reveal a great deal about the harm that negative emotions deliver to the heart, they also clearly demonstrate the amazing healing power of positive emotions. In my 25 years as a cardiologist performing clinical trials and treating patients, I've seen firsthand how we can harness optimism, confidence, laughter, social connections, and relaxation to help our hearts get and stay healthy.

5. Laugh Hysterically

Deep belly laughter triggers the release of endorphins, which activate receptors in our blood vessels' linings that signal the production of nitric oxide. This powerful chemical causes blood vessel dilation, increases blood flow, reduces vascular inflammation and buildup of cholesterol plaque, and decreases platelet stickiness, which lowers the risk of blood clots.

6. In an early study, my team saw that people with heart disease were 40 percent less likely to use humor in an uncomfortable situation, such as having a waiter spill a drink on them, than people with healthy hearts. In another study, when we asked people to watch a clip from Saving Private Ryan or There's Something About Mary, we found that participants' blood vessels were narrowing by up to 50 percent during the stress-inducing clip, while vessel dilation in people who watched a funny clip increased 22 percent. After just 15 minutes of laughing, volunteers got the same vascular benefit as they would from spending 15 to 30 minutes at the gym or taking a daily statin.

7. Listen to the Music

Medical science is now proving what people have known for hundreds of years: that music is deeply healing. In one study, researchers found that listening to music 25 minutes daily for four weeks resulted in a 12 mm Hg reduction in systolic blood pressure (the top number) and a 5 mm Hg decrease in diastolic blood pressure (the bottom number). Results like these are equivalent to the benefit of taking a strong blood pressure medication. The calming effect of music is so powerful that listening to relaxing music before cardiac surgery was more effective at reducing stress than a sedative medication.

8. And a group who listened to music after surgery fared better than patients who received the sedative. One theory is that music acts directly on the body's autonomic nervous system, which is responsible for heart rate and blood pressure.

9. Cuddle Up

During childhood visits to the doctor, I remember feeling that everything would be fine when my pediatrician would place his hand on my upper shoulder as he listened to my lungs. Early in my training, I did the same thing to my patients. Several studies support the idea that interpersonal touch has important heart-health benefits. In one study, women who received frequent hugs from their partner showed reduced heart rates and blood pressure as well as higher levels of the powerful neurotransmitter oxytocin, which leads to blood vessel dilation.

Language Enhancement

1. The boldfaced words in the following sentences are collocations of various types, for example, preposition + noun; verb + noun; noun + preposition, adjective + preposition.

A. Complete each collocation with the exact word from Reading 1.

1) ...with a very somber look, I inform patients that … (Para. 1)
2) ...one thing they absolutely must do in order to make a successful recovery after a cardiac event
3) … stress, and an inability to deal with it, is a direct contributor to heart disease. (Para. 2)
4) ... a study … found that anxiety was associated with a 26 percent increase … (Para. 2)
5) Anger and hostility rank at the top of the list of heart-harmful emotions. (Para. 3)
6) … patients who suffered a heart attack reported significant anger within the previous year, …
7) … they also clearly demonstrate the amazing healing power of positive emotions. (Para. 4)
8) In my 25 years as a cardiologist performing clinical trials and treating patients, … (Para. 4)
9) Deep belly laughter triggers the release of endorphins, … (Para. 5)
10) ... which lowers the risk of blood clots. (Para. 5)
11) ... listening to music … resulted in a 12 mm Hg reduction in systolic blood pressure .. (Para. 7)
12) Results like these are equivalent to the benefit of taking a strong blood pressure medication.
13) listening to relaxing music ... was more effective at reducing stress than a sedative medication
14) … music acts directly on the body's autonomic nervous system, … (Para. 8)

B. Fill in the blanks with the collocations from Step A. Change the form where necessary.

1) Scientists have proved that moderate exercise triggers the release of chemicals in the brain that make you feel better.
2) Edward made a successful recovery from injuries and was expected to return to the competition next week.
3) The country has been ranking at the top of the list of the safest countries in the world for three consecutive (连续不断的) years.
4) Previous research has proven that being optimistic boosts overall physical health and lowers the risk of death from all causes.
5) With a very somber look, he asked the employees, "who is responsible for this inferior product?"
6) The anti-depression medication can act directly on the brain to change aspects of mental activity.
7) Nuclear explosions are measured in kilotons. Their energy release is equivalent to that of thousands of tons of TNT (炸药).
8) The great increase in tourism in recent years has resulted in a corresponding boom in hotel construction.

9）Numerous cases have demonstrated the amazing healing power of this very simple yoga pose.

10）Drinking very hot tea (at least 70° C) is associated with an eightfold increased risk of esophagus cancer (食道癌) compared with warm tea.

2. Study the word building skill in the box and work on the tasks.

A. There are some analogical words with their origins in the following table. Translate them into Chinese. The first one has been done as an example. Consult the dictionary where necessary.

Origin	Analogical word	Chinese meaning
white-collar / blue-collar	1) gold-collar	金领的（高级管理者的）
	2) gray-collar	灰领的（服务行业职工的）
	3) pink-collar	粉领的（女性上班族；从事主要由女性所任工作的）
landscape	4) seascape	海景
	5) moonscape	月面风景
baby-sit	6) house-sit	代为照看房子
high-rise	7) low-rise	低层建筑
overproduce	8) underproduce	生产不足；减产
broadcast	9) narrowcast	小范围播送

B. The boldfaced words in the following sentences are analogical words. Translate them into Chinese and figure out their origins. Consult the dictionary where necessary.

Analogical coinage	Chinese meaning	Origin
1) reader-friendly	一般人可以读懂的；通俗易懂的	**user-friendly**
2) workaholic	工作狂的	**alcoholic**
3) netizen	网民	**citizen**
4) underpopulation	人口稀少	**overpopulation**
5) animal rights	动物权利	**human rights**

3. Translating

中医有五千多年的历史，是中国古代劳动人民几千年来对抗疾病的经验总结。中医将人体看成是气、形、神的统一体，以“望、闻、问、切”为其独特的诊断过程。中医使用中药、针灸以及许多其他治疗手段，使人体达到阴阳调和。阴阳和五行是中医的理论基础。五行是自然界中的五种基本物质，即金、木、水、火、土。

Traditional Chinese Medicine (TCM) has a long history of more than 5,000 years. It is a collection of the experience of working people over many centuries of struggle against diseases. TCM considers the human body as a unity of QI, XING and SHEN. The diagnostic process of TCM distinguishes itself by “observation / inspection, auscultation and olfaction, inquiry and pulse diagnosis”. TCM employs Chinese herbal medicine, acupuncture and many other treatments to make the human body harmonious between YIN and YANG. The concepts of YIN-YANG and WU XING laid a theoretical foundation for TCM. WU XING refers to five basic substances in Nature, namely, metal, wood, water, fire, and earth.

Unit 4
Reading 1

Why Your Cell Phone Sounds Female

1 Ask Siri if she's a woman. Go ahead; try it. She'll tell you she's genderless. "Like cacti. And certain species of fish," she might say. So is Amazon's Alexa, Microsoft's Cortana, Samsung's S Voice, and Google Now. But, man, do they ever sound a lot like women. Culturally, we think of them as ladies too. (In Old Norse, Siri translates to a beautiful woman who leads you to victory.") We assign female pronouns to them, and, in turn, they fold feminine turns of phrase into their robotic and occasionally inane answers to our requests.

2 If we prize gender diversity in other areas of daily life, why does our technology sound so female?

3 It would be easy to credit — or fault — male designers, perhaps influenced by science fiction. (Interesting fact: In the original Star Trek TV series, the voice of the Federation's onboard computers was supplied by creator Gene Roddenberry's wife, Majel Barrett.)

4 But the biggest reason for the female phone fixation rests in social science. "Research indicates there's likely to be greater acceptance of female speech," says Karl MacDorman, an associate professor at Indiana University who specializes in human-computer interaction. MacDorman and his team played clips of male and female voices to people of both genders, then asked them to identify which they preferred. The researchers also measured the way participants responded to the voices. In a 2011 paper, they reported that both women and men said female voices came across as warmer. In practice, women even showed a subconscious preference for responding to females; men remained subconsciously neutral.

5 Why the bias? Stanford University communications professor Clifford Nass, who coauthored the field's seminal book, Wired for Speech, wrote that people tend to perceive female voices as helping them solve their problems by themselves, while they view male voices as authority figures who tell them the answers to their problems. We want technology to help us, but we also want to be the boss of it, so we are more likely to opt for a female interface.

6 This inclination suggests that companies will make a better impression on a broader group of customers with a woman's voice. But not just any voice. It has to align with a brand's personality. For help with that, companies often turn to Greg Pal, vice president of marketing, strategy, and business development at Nuance Communications, which licenses its library of more than 100 voices. Pal insists that some brands choose male speakers. He turned on his iPhone and pulled up the Domino's Pizza app, which has an assistant, Dom. He sounded like my high school English teacher — educated and helpful but not overbearing. That's about right for a brand attempting to appeal to guys ordering pies before the big game.

7 As voice technology improves, though, designers say diversity will too. Many devices already let you customize a voice interface. Homer Simpson can tell you where to take a left on your GPS device. And Siri can become a sir, if you take the time to reprogram. Want to know how to do it? Ask her. She'll tell you in her uniquely warm, helpful — and female — tone.

Language Enhancement

1. The words in the box below are either active words or academic words marked in the Academic Word List (AWL). Choose the most suitable one to complete each sentence.

assign	creator	credit	diversity	fault
inclination	license	neutral	prize	tone

1) After winning the championship, the team's players all credited / credit their coach for helping them succeed.
2) All of the options offered here are reasonable ones; therefore, no one could be faulted for picking one over the other.
3) In writing to inquire into the matter, he kept the tone of the letter formal and businesslike.
4) Scientists studying the species of marine life said the diversity of life in the deep sea was much greater than previously believed.
5) The food company prizes their good reputation and always places top priority on food security.
6) We've assigned Chris to the advertising campaign and Steve to the marketing.
7) Both sides have agreed to attend a peace conference on the condition that it is held in a neutral country.
8) My initial inclination was to say no to their terms, but I finally decided to give in and continued the negotiation.
9) When a daycare center is licensed you can be assured at least to some degree that the center itself is inspected with some regularity.
10) Being the creator of a successful cartoon series, he has been an idol among children through the ages.

2. A preposition can be very useful in forming a collocation.

A. Scan Reading 1 and fill in the blanks with the prepositions used in the collocations.

1) But the biggest reason for the female phone fixation rests in social science. (Para. 4)
2) … an associate professor … who specializes in human-computer interaction. (Para. 4)
3) … both women and men said female voices came across as warmer. (Para. 4)
4) In practice, women even showed a subconscious preference for responding to females (Para. 4)
5) … we are more likely to opt for a female interface. (Para. 5)
6) … companies will make a better impression on a broader group of customers … (Para. 6)
7) It has to align with a brand's personality. (Para. 6)
8) For help with that, companies often turn to Greg Pal, … (Para. 6)
9) That's about right for a brand attempting to appeal to guys ordering pies ... (Para. 6)

B. Complete the sentences with the collocations from Step A. Change the form where necessary.

1) Rural things always appeal to me. I like the fresh air, the country roads, even the smells of the soil.
2) The problem of mismanagement rests in the fact that they do not have a competent leader.
3) After I did my engineering degree, I began to specialize in computer technology and I even had my own company for a while.
4) If you opt for something, you choose it or decide to do it in preference to anything else.
5) Urban residents often picture rural life as leisurely and comfortable, but in practice, that may not be the case.

4. Study the word building skill in the box and work on the tasks below.

A. Match the English native words with their synonyms in the Loan Words column.

English Native Words	Loan Words
f 1) Good	a) collection
d 2) broad-minded	b) sacred
j 3) Cling	c) suppress
a 4) Cluster	d) liberal
e 5) Gloomy	e) depressed
i 6) Hinder	f) beneficial
b 7) Holy	g) spirit
g 8) Mood	h) figure
h 9) Shape	i) obstruct
c 10) withhold	j) adhere

B. The underlined words in the following sentences are from Reading 1. Choose the synonym of each of them from the given choices.

b 1) You can't appreciate how much I **prize** the friendship between us
a) judge b) value c) award

a 2) It is very essential to maintain cultural **diversity** and integrity in a healthy community.
a) variety b) similarity c) unity

c 3) The invention of the industrial robot is **credited** to this enterprising team.
a) praised b) directed c) attributed

c 4) Urban planners **in practice** have to take account of many interest groups in society.
a) with possibility b) in conclusion c) in reality

b 5) The report to the general conference should be a **neutral** one and be based on facts.
a) prejudiced b) unbiased c) mature

a 6) Optimism, as an essential personality trait, is an individual's cognitive and emotional **inclination** toward the current and future condition.
a) tendency b) involvement c) opinion

c 7) The State Department spokesman said that the domestic oil price has been **aligned** with that of the world market.
a) cooperated b) accompanied c) coordinated

a 8) The whole **tone** of the president's speech was one of continuity and stability.
a) expression b) pitch c) attitude

5. Translate

在组成经济合作与发展组织的富裕和中等收入国家，全职女性的平均工资是男性的 85%。这并不是因为雇主付给女性的薪水要低于男性。一家咨询公司收集的 25 个国家的数据显示，女性在同一份工作上的收入是男性的 98%。真正的原因有两点。第一，在薪水较低、晋升机会不大的岗位上，女性的数量超过男性。第二，男女在工作及行业方面被隔离；而那些女性占主导地位的工作及行业薪酬更低。

In the rich and middle-income countries that make up the OECD, the average wage of a woman working full-time is 85% that of a man. This is not because employers pay a woman less than they would have paid a man in her place. Data from 25 countries collected by a consultancy show that women earn 98% as much as men who do the same job for the same employer. The real reason is twofold. Firstly, women outnumber men in positions with lower salaries and little chance of promotion; secondly, men and women are segregated in occupations and industries, and those where women predominate are paid less.

Unit 5
Reading 1

Working with Robots: Human and Machine Coexistence in the Workforce

1 The pervasive fear that artificial intelligence (AI) will take over human economic livelihood has been felt in places like the manufacturing sector, as large areas of the industry automate labor formerly done by humans. However, proponents of machine learning say ultimately AI and robotics will improve the way we do virtually everything, and ultimately create new jobs.

2 Still, nearly 40 percent of U.S. jobs were slated as a "high risk" for automation by the early 2030s in a March 2017 report by PricewaterhouseCoopers (PwC). While the PwC report acknowledges it's unlikely all those jobs will be automated for "a variety of economic, legal, and regulatory reasons," PwC also acknowledges that new tech typically means the creation of new jobs for human workers as well, conceding "the net impact of automation on total employment is therefore unclear."

3 Many technologists claim that the new job creation will offset some of the pain of displacement; retraining programs and continuing education opportunities are key to bringing in displaced workers into the new high-tech fold.

4 "Ever since the industrial revolution, we've created technology that in theory has displaced workers, and yet growth continues," Chris Volinsky, assistant Vice President of inventive science at AT&T Labs, told Business News Daily. "It's a displacement of work from more tasks to those tasks which require more education and more technology, so work gets displaced, but workers are constantly evolving and being retrained."

5 However, the pace of technological growth is so fast that many workers might not find this a truly viable option, said Moshe Vardi, professor of computer science at Rice University and fellow at the Institute of Electrical and Electronics Engineers.

6 "As AI becomes more effective and complex, the zone of 'automated jobs' will continue to widen across industries and verticals," Vardi said. "Workers are racing against the machines, and to stay ahead of the game, they need to be willing to continually refresh and upgrade their skills. The jobs least likely to be automated are those that combine nonroutine technical skills in combination with people skills."

7 What's on the horizon?

The pace of change has workers both excited and anxious about what the future holds. According to a survey conducted by Atlassian, 87 percent of respondents expect AI to change their jobs by 2020, with 76 percent responding that some or half of their job could be performed by an algorithm or robot. And while 64 percent said they trust AI's ability to properly complete a task, 80 percent are concerned about a subsequent spike in unemployment.

8. "If harnessed correctly, AI can become our team's 'sixth man', moving beyond digital assistants and chatbots, and freeing up time and brains for us to tackle society's most complex problems," Atlassian's report reads.

9. "AI is, first and foremost, a tool that makes humans more productive, not unlike a hammer or a steam shovel," said Manuel Ebert, founder of AI and machine intelligence consultancy summer.ai. "If one human can produce more in the same time, that means we need fewer humans to satisfy the same demand. That is where displacement comes from."

10. However, Ebert continued, when productivity increases and costs decrease, often the demand for those goods and services increases and helps drive the creation of new jobs.

11 "Think the printing press and books or the assembly line and cars," he said. "So, the interesting question is where can AI create demand for things that were previously (prohibitively) expensive?"

12 The PwC report anticipates a rise in average pretax incomes because of mass adoption but acknowledges "these benefits may not be evenly spread across income groups."

13 "There is therefore a case for some form of government intervention to ensure that the potential gains from automation are shared more widely across society through policies like increased investment in vocational education and training," the report reads.

14 Others have suggested a universal basic income of some kind, which would essentially offer payments to citizens that could cover necessities like groceries or rent and mortgage payments.

15 **How will the job market transition into automation?**

While it is generally agreed that some steps need to be taken to relieve the pain of transition, AI proponents like Volinsky argue that the benefits of these technologies far outweigh the negatives. For example, he said, AT&T is utilizing drones and machine learning to accelerate inspection and maintenance of cell towers. Instead of sending a worker up, the company now flies drones to inspect the antennae.

16 "(The drone) flies up with HD video and sends footage back to a technician on the ground to inspect," Volinsky said. "It might take a half hour to do a full detailed inspection of one of those towers, even though the technician is only interested in certain parts of that video."

17 That's where AI comes in: Machine learning can be used to identify potential problem areas and highlight key points of interest the human technician needs to analyze, Volinsky said. By doing so, it can reduce the half hour task to a matter of minutes, removing the technician's need to scan through useless pieces of video to find the value.

18 AI is also making headway in customer service, internal decision-making and the way companies track their customer relationships, to name a few examples. Each of these inroads represent only the beginning of the AI revolution, Volinsky said, and these tools will be essential as they proliferate.

19 "I like to think of AI as taking the dull and routine parts out of people's work and helping humans focus on their real expertise, which is identifying problems and focusing on what else needs to be done," Volinsky said.

20 As AI more prominently enters the workforce, humans will need to prepare for continued waves of automation by learning new skills and adapting to a changing economy, while harnessing the capabilities of AI to solve problems that were previously out of reach. A symbiotic relationship between man and machine, then, appears far more desirable than a war for prominence.

Language Enhancement

1. **The words in the box below are either active words or academic words marked in the Academic Word List (AWL). Choose the most suitable one to complete each sentence. Change the form where necessary.**

automate	automation	displace	humble	maintenance	offset
regulatory	subsequent	ultimately	upgrade	utilize	virtually

1) The continuous rainfall and the subsequent flooding have thrown city traffic into chaos.
2) Despite our heroic efforts to save the business, it ultimately went bankrupt.
3) In the information era, communications with friends far-away can be virtually simultaneous.
4) To boost productivity, the company plans to introduce robots to automate the process that used to be manual.
5) Many of the company's workers were displaced by robots as a result of the rapid development of artificial intelligence.
6) Due to the severe misconduct in the crisis management, he was transferred to a humble position for the time being.
7) The wind was so violent that our maintenance staff could not get out to fix the antennae (天线) until a few hours later.
8) Across the company, robust Asian demand almost entirely offset falling sales of jewelry and watches in European and American markets.
9) Word processing has led to the advancement of office automation and is definitely the best-established technology in the office today.
10) In a notice published on its website today, Baidu said it had deleted indecent content and links and improved its regulatory system.
11) This training program will help workers to upgrade their computer skills and be better prepared for the automated process.
12) The websites you visit on a regular basis utilize psychology to understand how people read, use and interpret online information.

2. Complete the sentences with phrases from Reading 1. Their meanings and paragraph numbers have been given in brackets.

1) During the first few months after he took over the loss-making business, he lived on a miserable schedule, sleeping only five hours a day. (gained control over; Para. 1)
2) I agree with you in theory, but realistically I don't think we have enough funding to accomplish that. (theoretically; ideally; Para. 4)
3) Scientists believe that more breakthroughs in many other Information Technology domains are on the horizon. (coming in the near future; Para. 7)
4) Flexible working arrangements may cut down on your stress as well as free up some of your time. (make available; Para. 8)
5) What the consumers need first and foremost from food companies is food quality and safety. (above all else; Para. 9)
6) The factory needs someone to handle the debt problem and that is where you come in. (have a particular role; Para. 17)
7) While the research might be biological, the ability to make headway on this centuries-old problem is technological. (make progress; Para. 18)

3.Word building

A. Classify the following pairs of antonyms.

1) Gradable antonymy: hot — cold, old — young, love — hate, rich — poor

2) Complementary antonymy: present — absent, male — female, single — married, on — off

3) Relational antonymy: husband — wife, buy — sell, above — below, give — receive

B. The underlined words in the following sentences are either original words or derived words from Reading 1. Choose the antonym of each of them from the given choices.

a 1) Ambiguity (歧义), as a common and pervasive language phenomenon, has long been a major concern of many Chinese and foreign scholars.

a) rare b) unpopular c) influential

c 2) The experiment results show that this algorithm could deal with automatic human face detecting and tracking efficiently.

a) intellectual b) fixed c) manual

b 3) The net conclusion of the research is that immediate actions must be taken to save the environment.

a) gross b) initial c) final

b 4) They must be confident enough to know when they have found a scientific truth, but humble enough to admit when they are wrong.

a) polite b) superior c) advanced

a 5) There may come a time when our reliance on advertising revenues to pay for this journalism is no longer a viable business model.

a) impossible b) difficult c) invariable

c 6) As the products are to be manufactured on this new robotic assembly line, one of the foremost considerations must be quality of the products.

a) indirect b) primary c) secondary

a 7) In recent years, commercial, cultural, travel and other contacts have proliferated between Europe and China.

a) shrunk b) grown c) maintained

b 8) The report was given a prominent position on the front page.

a) outstanding b) invisible c) covered

4.Translate

人工智能是由美国计算机科学家约翰·麦卡锡（John McCarthy）于 1956 年在达特茅斯会议（The Dartmouth Conference）上创造的。今天，它是一个涵盖了从机器人过程自动化到实际机器人技术的包罗万象的术语（umbrella term）。最近，它的知名度越来越高，部分原因是大数据，也就是企业收集数据，在速度、规模和种类等方面的增长。人工智能可以比人类更有效地执行数据识别模式等任务，从而使企业能够从数据中获得更多的灵感。

Artificial Intelligence (AI) was coined by John McCarthy, an American computer scientist, in 1956 at The Dartmouth Conference. Today, it is an umbrella term that includes everything from robotic process automation to actual robotics. It has gained prominence recently due, in part, to big data, or the increase in speed, size and variety of data that businesses are now collecting. AI can perform tasks such as identifying patterns in the data more efficiently than humans, enabling businesses to gain more insight out of their data.

Unit 6
Reading 1

You Talk Too Much!

1 We had a couple over for dinner the other night — friends of friends — and at first, the woman seemed charming.

2 Then she seemed to be sort of monopolizing the conversation.

2 Then we realized she NEVER SHUT UP.

4 And then we noticed her husband's head cocked at a weird angle. He had fallen asleep at the dinner table.

1 Lucky guy.

2 What's shocking is not that some people talk too much. What's shocking is that they don't seem to pick up on even the most obvious cues — a passed-out spouse, for instance.

3 It turns out there are two basic types of over-talkers. The first actually work at being entertaining because they feed off our appreciation. Whether they succeed is up to the listener. The second group is made up of those who fear that if you stop listening, they stop living. It's a problem psychologists ascribe to everything from loneliness to insecurity to arrogance.

4 University of Texas professor of psychology Art Markman, author of the book Smart Change, says nonstop talkers "need social interaction to survive, so they're just looking to plug into somebody and don't care who it is. They smell social interaction and go into a feeding frenzy."

5 One neighbor of ours is such an extreme yakaholic. Everyone in the area dives inside when they see her coming, lest she catch their eye and start saying, "Our grandson was just given a promotion, and you know how you can get a promotion at his job? The only way you can get a promotion is ..." This is not chitchat. This is an act of aggression. Talkers mug listeners — they steal their time.

6 If you're ever confronted by a yakker — and you will be — try these coping strategies:

7 At Work

To fend off chattering coworkers, says Jennifer Kalita, CEO of the Vesta Group, a communications consulting firm in Washington, DC, "add the expression hard stop to your vocabulary. At the beginning of a meeting, say 'I have a hard stop at 3 p.m., so let's dig right in.'" Somehow a hard stop sounds set in stone, giving you the perfect out.

8 Another idea is to keep a talkative person on track by playing dumb, says psychotherapist LeslieBeth Wish, author of Smart Relationships. "Say 'I like what you're saying; I just want to make sure I've got it right.' " This way, you force the talker to focus.

13 **With Friends**

If a longtime friend routinely goes into talk radio mode but you don't want to sever the relationship, "plan activities where talking isn't allowed," says Kalita. "Go to the movies instead of dinner. Attend a workshop together instead of a party. Take a Zumba class instead of lunch."

14 **At Home**

"My mom and I would watch TV while Dad sat in his recliner and talked," recalls writer Jess Kennedy Williams. "We learned to block him out until his voice went up like he was asking a question, and then we'd say, 'Yes, I know,' or whatever."

15 There are far better ways to keep peace in the family. The best one is to simply keep yourself busy while the talkers talk so you won't feel they're totally wasting your time. Sort the laundry, paint the kitchen, brand the cattle. Multitasking is key.

16 **A Yakker? Moi**

Might you be an unwitting member of the chattering classes? To find out, says Kalita, examine your chatting habits like a detective. "When your friend is speaking, are you really listening or just thinking about the thing you're going to say next? When your friend tells you a story about an alligator, do you need to tell a bigger, more shocking story about an alligator?"

17 Perhaps most saliently: Did you ask any questions? Did you follow up with more questions? And were these questions not of the "That's awful. Wanna hear what happened to me today?" variety? A real conversation involves listening, asking, nodding, looking surprised — basically, all the stuff you see Dr. Phil doing when he's not telling his guests how crazy they are.

18 There is, however, one time when you are absolutely allowed to dominate the conversation to your heart's content , and that is when that heart of yours is bursting. If your dog died, your daughter got engaged, you just lost your job or got elected to the Oval Office — if it's a really big moment in your life — babble on.

19. Just don't forget to thank whoever's listening for his or her time.

Language Enhancement

1. The italicized words and expressions in the box have more than one meaning. Decide which meaning is suitable for the sentence.

a 1) The soldier cocked the gun, ready to shoot.

b 2) He cocked his head to one side and fell asleep.

a. pull back the trigger of …

b. turn … (a part of one's body) so that it is at an angle

b 3) She was nervous, but no one picked up on it.

a 4) The speaker picked up on the last comment.

a. continue talking about b. notice or become aware of

a 5) She feels she is up to the challenges in the new job.

b 6) The amount of donation is entirely up to the donors.

a. capable of dealing with b. decided by

b 7) The teammates are able to feed off each other's ideas.

a 8) These bears feed off the leaves of this special tree.

a. get food from b. gain support from

a 9) Please plug the scanner into the computer.

b 10) The computer program takes data from these sources and automatically plugs it into the equation.

a. connect … to

b. put (information) into …

a 11) He just grabbed a plate and dug in.

b 12) Let's dig in and focus on the projects right now.

a. begin eating

b. begin work with great determination

b 13) Try to block out the distractions and dig into your work.

a 14) We put on music to block out the sound of the traffic.

a. stop (noise) from coming in

b. ignore

a 15) They no longer brand their cattle.

b 16) They branded him a coward.

a. put a mark on (an animal) to show who owns it

b. describe with a word that expresses strong criticism

b 17) He followed up his early findings with another study.

a 18) The references in the book were too vague to follow up.

a. try to get more information about

b. add to (sth. that one has just done) by doing sth. else

2. Study the word building skill in the box and work on the tasks below.

A. The original meaning of the word "power" is "ability", and then it has gained more meanings in the process of radiation. Several meanings (a–j) of it are listed below. Read the following sentences (1–6) and put the letters (a–j) into the brackets after each sentence.

a) the ability to do or act; particular faculty of body or mind (primary meaning)

b) the ability to control people or things

c) the authority of a person or group to do sth.

d) a country with a lot of influence in world affairs, or with great military strength

e) the physical strength or effect of sth.

f) the public supply of electricity

g) energy that can be collected and used to operate a machine, to make electricity

h) the number of times that an amount is to be multiplied by itself (乘方；幂)

i) the amount by which a lens can make objects appear larger (放大倍率)

j) a good or evil spirit that controls the lives of others

1) The power of the explosion smashed windows across the street. (e)

2) People should have more power over the decisions that affect their lives. (b)

3) A writer's power of observation is the decisive factor of a successful novel. (a)

4) The police have been given special powers to help them in the fight against terrorism. (c)

5) Why did the United States become the leading economic power of the 20th century? (d)

6) The scientist said the facilities appeared to be designed primarily for civilian nuclear power. (g)

B. The underlined words in the following sentences are from Reading 1, and they all have more than one meaning. Choose the suitable meaning for each of them from the given choices.

a 1) This organization is trying to monopolize the supply of oil in this region.

a) to have or take control of the largest part of sth., so that other people are prevented from sharing it.

b) to take a large part of sb.'s attention or time so that they are unable to speak to or deal with other people

b 2) They ascribed great importance to these polices which were made to boost the local economy.

a) to consider that sth. is caused by a particular thing or person

b) to consider that sb. / sth. has or should have a particular quality

b 3) He said that this country had no history of military aggression and expansion and needs a peaceful and steady international environment.

a) feelings of anger and hatred that may result in threatening or violent behavior

b) a violent attack or threats by one person against another person or by one country against another country

a 4) Due to the poor performance of the company, the CEO found himself confronted by massive opposition.

a) to appear and need to be dealt with by sb.

b) to face sb. so that they cannot avoid seeing and hearing you

b 5) That would allow us to sever all our ties from the company and any possible conflict would be eliminated.

a) to cut sth. into two pieces; to cut sth. off sth.

b) to completely end a relationship or all communication with sb.

b 6) The train crash that caused heavy casualties dominated the news.

a) to control or have a lot of influence over sb. / sth., especially in an unpleasant way

b) to be the most important or noticeable feature of sth.

3. Translate

Translate into English the following paragraph, paying attention to the translation of set phrases.

关于销售技巧的早期书籍包含了诸如道德、服务、人际关系以及努力工作等关键词。这些都促成了一个概念：销售人员应与顾客建立友谊。后来，挨家挨户推销的销售人员发现他们可以通过使用特定的词语和特定的说服方法来增加销售额，这促成了销售技巧的完善和传播(proliferation)。这些技巧的重点不是顾客的需求，也不是建立友谊，而是通过成交技术和方法（closing techniques and methods）来完成一次性销售。

Early books about sales techniques included key words like ethics, service, relationships, and hard work. These all led to the idea of building a friendship with your customers. Later, door-to-door salesmen discovered that they could increase their sales by using specific words and specific persuasion methods. This led to the perfecting and proliferation of sales techniques that focused not on the customer's needs or building a relationship, but on closing techniques and methods that helped complete a one-time sale.